

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 04
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

Title: Design and Analysis of Pre-emphasis and De-emphasis Networks for FM Systems
Objectives:

1. To design pre-emphasis (high-pass) and de-emphasis (low-pass) filter
2. To analyze the magnitude and phase response of the networks
3. To observe the effect of the time constant (τ) on the frequency response

OUTCOME:

1. analyze and compare analog modulation schemes.
2. understand working of analog communication transmitter and receiver systems
3. understand noise effect on the performance of communication systems.

• **Theory:**

In FM, noise power increases linearly with frequency (triangular noise spectrum). Since most audio signals have less energy at high frequencies, the high-frequency part of the message is easily buried in noise.

- **Pre-emphasis:** A high-pass filter is used at the transmitter to boost high frequencies before noise is added. The standard time constant is $\tau = RC = 75\mu s$ (or $50\mu s$ in some regions).
- **De-emphasis:** A low-pass filter at the receiver attenuates the high frequencies (and the noise) back to their original levels, restoring the original balance of the signal.

$$H_{pre}(f) = 1 + j(f/f_c)$$

$$H_{de}(f) = \frac{1}{1 + j(f/f_c)}$$

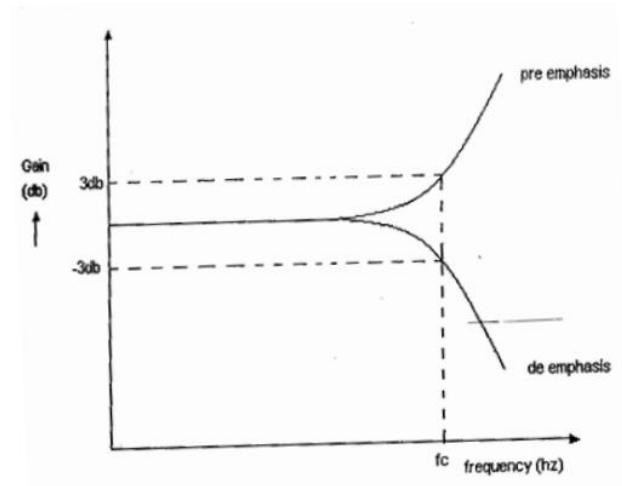
Where the cut-off frequency is $f_c = \frac{1}{2\pi RC}$

• **Design Parameters:**

- Time constant (τ) = 75×10^{-6} seconds
- Cut-off Frequency (f_c) ≈ 2.122 KHz
- Sampling Frequency (f_s) = 44.1 KHz (standard audio)

• **Procedure:**

- Open MATLAB Online or desktop version.
- Create a new script and copy the provided code.
- Calculate the theoretical f_c using the formula $f_c = 1/(2\pi\tau)$
- Develop a logic and program to see the frequency response
- Observe the 3 dB point on the plots to verify if it matches the calculated f_c
- Observe the "Combined Response" plot to ensure the net gain is 0 dB (flat).



- **Variations in Parameters:**

- Change τ : Change $\tau=50\mu\text{s}$. Observe the change and record
- Square wave input: Pass a square wave through pre-emphasis filter. You will see “overshoot” or spikes at the transition, representing the high-frequency boost.

- **Post lab activity:**

- What is the effect of noise in FM
- What conclusions can be drawn from a noise triangle?
- In your opinion, FM advantageous or disadvantageous over AM? Justify.

Signature of faculty in-charge